

The Teammate Divide: How Humans Approach Teamwork Differently with Human and AI Teammates

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As Artificial Intelligence (AI) technologies begin to become more fundamental to society, their design and integration as teammates that work alongside humans will become commonplace. Recently, research into these human-AI teams has shown that the perceptions humans form of the capabilities and goals of their AI and human teammates differ. However, while humans perceive their teammates differently, it is still unclear whether these perceptual differences extend to the actual teamwork humans have with their teammates. To address this gap, this article reports on a mixed-methods research study of 32 human-AI teams with 2 human teammates and 1 AI teammate. Human and AI teammates occupied similar roles and were tasked with completing a team task in an esports environment. Based on this team task, participants rated the quality of teamwork they experienced with their human and AI teammate, and an interview was conducted to identify why these perceptions may have developed during the team task. Quantitative results show that when working with both another human and an AI teammate, participants perceived a stronger level of teamwork (i.e., interdependence, information certainty, and mutual dependence) and less conflict with their other human teammate compared to their AI teammate. Additionally, qualitative insights highlight that these perceptual differences were formed due to participants having greater expectations for their AI teammates, which may not have been met, and finding it easier to form a strong mental model of their human teammate compared to their AI teammate. Based on these results, this article discusses the potential downstream impacts that teamwork differences could have on human-AI teams and how to address the root causes of these differences to improve team effectiveness.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in collaborative and social computing**; • **Computing methodologies** → **Artificial intelligence**.

Additional Key Words and Phrases: Human-AI Teams, Artificial Intelligence, Teamwork, Teammate Artificiality, Team Effectiveness, Mixed-Methods

ACM Reference Format:

Christopher Flathmann, Han Nguyen, Kwame Andre, and Chase Guynup. 2026. The Teammate Divide: How Humans Approach Teamwork Differently with Human and AI Teammates. *Proc. ACM Hum.-Comput. Interact.* 10, 1, Article GROUP017 (March 2026), 25 pages. <https://doi.org/10.1145/3799452>

1 Introduction

With recent expansions in the application of artificial intelligence (AI), AI teammates can operate as interdependent collaborators and perform teamwork with humans [51, 83]. At a base level, these human-AI teams are composed of at least one human and one AI teammate and require the

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ACM 2573-0142/2026/3-ARTGROUP017

<https://doi.org/10.1145/3799452>

human and AI to perform interdependent roles to complete a shared task [52]. In human-AI teams, researchers define teammates as entities that operate with a high degree of autonomy, work toward a shared goal, and present adequate capabilities that allow them to independently contribute to that shared goal [51]. When teammates interact, their interactions create teamwork, which refers to a process where behavior, information, and understanding all exist interdependently between teammates and help teammates complete a shared goal [52]. In recent years, researchers have explored how teamwork and interdependence differ between human-human and human-AI teams, showing that human-AI teams will create distinct types of interdependence from human-human teams [44]. However, in human-AI teams, multiple human teammates and multiple AI teammates can exist, meaning that, by the definition of teamwork, human-AI teams can leverage both human-human and human-AI teamwork [2, 49]. In turn, it may not be enough to explore human-AI teaming as monolithic, as the human-human and human-AI teamwork present in human-AI teams might differ.

To lend credence to the idea that teamwork might differ, Prior research has repeatedly demonstrated distinct perceptions of AI and human teammates [45]. These observations have been present when human-human teams are compared to human-AI teams [43] and when human and AI teammates are compared simultaneously in human-AI teams [21]. Specifically, researchers have commonly explored trust and performance for both types of teammates, showing strong differences between how humans perceive the trust and performance of human and AI teammates [23]. However, these baseline perceptions fail to capture teamwork processes. As stated, teamwork involves teammates engaging in a shared process, and it is critical to understand if humans approach this process differently when working with AI teammates or other human teammates [22, 81]. In traditional teams, teammates express these processes through shared understanding, behavioral coordination, social influence, and mutual adaptation [61]. With research continuing to show that humans approach trust differently with AI and other humans [15], it is also important to understand if humans approach these teamwork processes differently. In regard to trust, Researchers found these differences imperative in guiding the design of modern AI teammates, which rely on explainability, transparency, and trust repair [15]. In a similar vein, should the design, as AI teammates would not be able to simply replicate human teammate behaviors, AI teammates would need to make up for losses in teamwork effectiveness, and human teammates would need to cope with the mental workload that these differences impose.

As such, this work has identified a gap within the human-AI teaming space. Current research does not understand whether humans perceive teamwork between themselves and their human teammates and themselves and their AI teammates to be different. Moreover, if these two types of teamwork are perceived to be different, it is unclear why these differences exist and how these differences will impact the design, integration, and performance of human-AI teams.

To remedy this gap, this article reports on a mixed-methods research study that explored how 32 human-AI teams, consisting of two human teammates and one AI teammate, developed unique understandings of interdependent human-human and human-AI teamwork processes. Specifically, we leveraged quantitative data to explore the perception of different teamwork processes and how they differed for both human and AI teammates. Additionally, we analyzed qualitative data to explore how participants differentiated between the teamwork they had with their AI teammate and their human teammate. In performing this study, we posed and answered the following two research questions:

RQ1: Do humans form distinct perceptions of teamwork for AI and human teammates when working in a human-AI team?

RQ2: How do humans approach teamwork differently between themselves, AI teammates, and other human teammates when working in a human-AI team?

This work makes three timely and important contributions to research surrounding human-AI teaming. First and foremost, this work will increase the ability for human-AI teams to balance the unique teamwork needs of human team members when interacting with their AI and human teammates. This contribution will become increasingly important as AI technology continues to be integrated alongside multiple human teammates. Second, this work will further expand upon what teamwork looks like in human-AI teams, which will increase the ability for AI teammates to measure and evaluate the effectiveness and needs of human-AI teams. Finally, this work will shed light on the potential deficiencies that hinder teamwork processes in human-AI teams, which will help improve the training and implementation of effective human-AI teams.

2 Background

Historically, the perception and evaluation of teamwork has explored the quality and strength of bonds and interactions that teammates form. Human-AI teams still lack in this area, so understanding what teamwork historically looks like and the current state of teamwork in human-AI teams is critical. As such, the following explores these two constructs.

2.1 Aspects that Define Strong Teamwork Processes

Historically, researchers have underpinned the core conceptualization of effective teamwork with the concept of interdependence [34]. Interdependence, in this context, refers to the degree to which team members rely on one another to complete their tasks and achieve shared goals, both structurally and behaviorally [40]. It is a relational construct, reflecting the reciprocal reliance among team members for resources, information, and expertise. Functionally, researchers describe interdependence through mechanisms such as information flow, task sequencing, and joint problem-solving [41]. For instance, pooled interdependence involves members contributing independently to a common goal, sequential interdependence implies a chain-like dependency, and reciprocal interdependence involves a more intricate, back-and-forth exchange among members [12]. Ultimately, interdependence is the critical differentiator distinguishing genuine teams from mere collections of individuals, as it mandates collaborative effort and mutual adjustment to effectively leverage diverse skills and perspectives [6]. When teams design interdependence effectively, it enhances performance by fostering shared mental models, reducing coordination costs, facilitating error correction, and increasing collective efficacy [52].

Strong interdependence involves more than merely dividing up tasks; it is a process where individual processes are combined and amplified to reach shared goals [52, 54]. Within team interdependence, researchers can define and explore multiple interdependent processes, such as continuous information sharing [11], the creation of shared understanding between teammates [48], behavioral coordination that drive interaction [42], providing mutual support and monitoring to teammates [61], and leveraging adaptive leadership to influence the behaviors of teammates [59, 61]. When these components of interdependence all manifest, a team's performance can far exceed what individuals or an unintegrated team could achieve [67]. This synergistic effect, where the collective output is greater than the sum of its individual contributions, is a key feature of high-performing teams [39, 67].

Creating these effective interdependence processes is not instantaneous; it often requires deliberate planning and ongoing adaptation. Traditionally, teams plan upfront to define roles, responsibilities, and communication protocols that foster interdependence [43]. During this planning phase, a critical consideration is ensuring that the capabilities and expertise of individual team members

are aligned with their assigned responsibilities, thereby maximizing the potential for effective collaboration [41, 66]. Similarly, a common approach involves reverse-engineering the shared team goal to systematically delineate the necessary roles and the specific interdependencies required to achieve that goal [67]. Researchers also recognize that some teams operate in dynamic contexts where interdependence is not solely pre-planned but evolves organically over time [52, 56]. Most notably, self-organizing teams exemplify this emergent development of interdependence [52]. In such teams, members may initially occupy more generalized or default roles, gradually developing specialized expertise and refining their interdependencies as they engage in joint tasks and learn from collective experiences [14, 43]. Regardless of the developmental trajectory, whether planned or emergent, the sustained success of interdependent teams hinges on the continuous awareness of each other's contributions and an unwavering commitment to their shared objective [43].

2.2 Human-AI Teams

Human-AI teams (HATs) represent an approach to human-AI collaboration where humans and AI agents work as teammates, engaging in interdependent tasks to accomplish a common goal [52]. Researchers have refined this definition to account for AI's increasing autonomy and agency. As such, HATs involve at least one human working cooperatively with at least one autonomous agent, with the AI possessing at least a partial degree of self-governance in decision-making, adaptation, and communication [44]. Furthermore, as AI systems become more agentic, performing dynamic functions in both teamwork and taskwork, the popularity of HATs is increasing [51]. However, for AI to operate effectively as teammates, AI agents must do more than simply execute tasks; they also need to engage with their human teammates in social and interpersonal ways. This holistic team performance requires a balance of both task-oriented and social-interpersonal capabilities. Thus, their success hinges on developing strong teamwork, utilizing robust social and performance capabilities rather than solely focusing on efficiency [58]. Consequently, HATs that excel in both areas demonstrate the most favorable views of their team processes [5].

Without a concerted focus on strong teamwork, Teams are unlikely to realize their full potential without strong teamwork, since the interplay of taskwork and teamwork is clear: teams high in both exhibit the most favorable views of their team processes [58]. However, developing strong teamwork in HATs presents a challenge due to the inherent differences between humans and AI. Humans excel at intuitive decision-making, emotional intelligence, and creativity [71]. In contrast, AI systems are adept at repetitive, high-volume, data-heavy tasks, offering speed and precision when dealing with large datasets [28]. These differences provide a clear motivation to leverage teamwork to accomplish shared tasks, but doing so proves difficult as it requires a shift from traditional teamwork [45]. For example, humans tend to view AI systems differently from other human teammates, often setting unrealistic expectations for AI teammates, which strains human-AI relationships and teamwork [83]. However, when these differences are overcome, human-AI teamwork offers significant benefits. For example, a study comparing clinicians diagnosing skin cancer with and without AI assistance reported that all clinicians showed improved performance when working with an AI system [35]. By acknowledging that these differences influence the overall performance of teams, it becomes critical to examine how they impact more specific team constructs and processes.

Despite human-AI teaming likely requiring a different approach to teamwork, Researchers have begun exploring how to model and form interdependent team processes in HATs [23]. For example, information certainty and shared understanding, which explore the ability for teammates to understand each other's intentions and thought processes, are critical for HATs, but their development is distinct due to the inherent differences between humans and AI [67, 69]. Humans must comprehend how their own features and those of AI systems can complement each other to

improve human-AI teamwork, but differences between these types of teammates complicate design and teamwork [33]. For example, when AI agents are tailored to match the personality and traits of human teammates, productivity and performance improve [29], and a similar improvement occurs when humans have accurate understandings of their AI teammates [83]. However, if there is a misalignment of expectations or understanding, lowering information certainty, then conflict is highly probable within human-AI teams. For example, when AI plays different roles in a teamwork setting, team members' preferences for how the AI should act can lead to conflicting behavior between human team members [8]. Outside of information certainty and shared understanding, researchers have begun exploring other interdependent processes in human-AI teams. Recent work highlighted that human-AI teams struggle to develop mutual dependence compared to human-human teammates due to challenges in responsibility negotiation [80]. For power and influence, humans often struggle to leverage social influence in human-AI teams due to expectations that AI teammates will not be receptive to said influence [18]. Coordination and behavioral interdependence are also complex processes, as the programmatic nature of AI systems often requires substantial overhead to enable complex coordination that allows humans and AI to adapt to one another [13].

Pulling these processes together, the development of interdependence in human-AI teams continues to be underdeveloped. However, where these works continue to fall short is that they explore interdependence, often individual interdependent processes, as a monolithic construct in human-AI teams. Rather, humans might develop distinct levels of interdependence between other humans and AI, so it is critical to explore how human-human and human-AI relationships might differentiate between these interdependence processes (i.e., information certainty, mutual dependence, power/influence, conflict, and behavioral interdependence). In other words, is it human-AI teams that find interdependence more difficult, or is it simply the human-AI relationships within human-AI teams that continue to struggle? Without this understanding, researchers will find it difficult to understand whether human-human or human-AI interdependence needs to be strengthened differently to create effective human-AI teams.

3 Methods

This work reports on an in-person research study that explored how humans perceive differences between human and AI teammates while simultaneously working with them in a human-AI team. The following details the design of this study and the data collected to explore the posed research questions.

3.1 Mixed-Methods Experimental Design

To maintain transparency and openness about this work, it is worth noting that the experiment reported on in this paper was part of a much larger experiment that explored a multitude of research questions. As part of that larger experiment, human participants worked in a human-AI team with varying numbers of AI teammates (1, 2, and 3). This article reports on a portion of data that was intentionally collected to explore the perceived differences between teamwork with human and AI teammates. As such, this article only reports data and statistical tests related to the empirical differences between the perception of AI teammates and the perception of human teammates. Further, to ensure this data is not conflated by the other experimental factors explored, the data used for this study only consists of interactions when two human participants were working with one AI teammate. This restriction allowed for the measurement of the perception participants formed for a single human teammate and a single AI teammate.

Given this scoping, this study explores an experimental design with a single independent variable: whether or not a perception measured was for an AI teammate or a human teammate. This experimental design was operationalized as a within-subjects variable, where participants rated

the perceived quality of their teammate with their AI and human teammate after working with them simultaneously in a team task. After this task, participants were asked to complete teamwork perception surveys for both their human and AI teammates. Additionally, participants completed a short interview after the entire experiment concluded, where they qualitatively compared the differences between their human and AI teammates. The result of this experimental design is a mixed-methods experiment that uses quantitative and qualitative data to measure and compare the perception of teamwork between human and AI teammates.

3.2 Experimental Task

This study leveraged the experimental task of Rocket League. Rocket League is an esports task that consists of teams of 2-4 playing a simulated and abstracted soccer-like game in vehicles. In the past three years, Rocket League has become a common research task to explore team processes in human-AI teams, such as social influence and communication [18, 21, 82]. Further, Rocket League has been explored as an ideal platform to design and research collaborative AI systems, as the sports-like proxy provides a natural environment to explore collaboration and train AI systems [62, 75]. For this study, collaborative AI teammates were selected from an open-source repository (RLBot) of custom-trained AI teammates that are able to account for and act around multiple human teammates. For this study, pilot tests were run, and an AI teammate was selected from the RLBot repository that closely matched the performance capabilities of pilot participants.

The benefits of using Rocket League for this specific study are that the capabilities and interactions of all teammates are identical, making interdependence easily comparable across teammates. Indeed, teammates in Rocket League all control a single car, interact in a shared space through the same actions, and communicate through their actions and constant awareness of one another (no verbal communication was allowed between human or AI participants in this study). In turn, this study was able to depict human and AI teammates that had similar roles, responsibilities, capabilities, and interaction modalities. As such, the only difference left between humans and AI would be the actual entity behind the car itself. Thus, this study was able to isolate the potential effects of perceiving a human or an AI teammate on the decisions made by each entity and participants' observations of them.

In relation to this study's concept of teamwork, which is heavily reliant on interdependence, Rocket League provides a straightforward platform to model and explore interdependence. Rocket League can allow for humans and AI teammates to play a simplified soccer task together, where a single shared objective of scoring goals exists. In this task, human and AI teammates work together to dribble a ball around a shared field, pass the ball between one another, shoot the ball toward an opposing team's goal, and continue this cycle for a predefined duration of time. This process requires mutual dependence and power, as a pass or blocked goal requires teammates to work around one another. Future interdependence is also critical, as teammates can continuously improve their coordination to score goals faster. Lastly, information certainty is fundamental, as the lack of direct communication means that teammates have to have a strong understanding of the intentions and capabilities of each other and adjust their own behavior accordingly.

Outside of this experiment, Rocket League provides important parallels and limitations to real-world teaming that scope this work's scope. Of note, these applications of Rocket League present two parallels to real-world teams: (1) self-organizing teams and (2) unmanned drone teams. For (1), teammates in this task are entirely responsible for self-organizing. Within the real world, self-organizing teams are common in software development, business consulting, and manufacturing [17, 26, 47]. These self-organizing teams are responsible for developing a strong level of interdependence through repeated interaction, as loosely or sometimes undefined roles require constant adaptation rather than forced interdependence through role definition. As such,

while Rocket League may not overlap task-wise with these teams, their underlying principles of self-organization would provide an opportunity to apply the results of this work. For (2), Rocket League presents a proxy for unmanned drone teams, where interaction does not directly happen through physical bodies but rather through vehicular proxies, where teammates have to account for physical resources, space, and collision. Outside of this experiment, drone-enabled teams are becoming commonplace in human-AI teaming, with the military, construction, manufacturing, and logistics all seeing drones as imperative to next-generation teamwork [1, 73]. As such, the spatial and physical constraints present in Rocket League provide an adequate proxy to the drone teams, allowing results to better translate outside of this synthetic task context. That said, despite the parallels that Rocket League shares with these two real-world team types, additional work will still be needed to extend the results of this work to these specific tasks. Further, the results of this work should also be reexplored outside of self-organizing and drone teams to understand how human-AI teams can develop interdependence and teamwork processes between humans, humans, and AI.

3.3 Participants

Due to using an esports platform with a low level of skill needed to play at a basic level, this study recruited a general population using a university subject pool. In total, 32 teams were included in this study, with two participants in each team for a total of 64 participants. This sample size is in line with human-AI teaming studies that recruit multiple human participants simultaneously [19, 45]. Participants had an average age of 18.34 (SD = 0.65). For gender, 15 participants identified as Male, 48 as Female, and 1 as Non-Binary. Race demographics were skewed toward white individuals, with 51 white, 1 black, 5 Latino or Hispanic, 3 Asian, and 4 Multicultural participants. Lastly, due to working with a university subject pool, the majority of students either had a high school degree (41) or completed some college (21), with a few students having completed associate's degrees (2). All participants completed a 1.75-hour research session, and they were compensated with course credit for their participation.

3.4 Procedure

The following procedure received IRB approval. As mentioned, the proposed work was part of a larger research study, so the following procedure will discuss elements not relevant to this study, but they have been included to maintain transparency and reproducibility.

Two participants were initially recruited through an online subject pool for a singular time slot, and a researcher waited for both participants to arrive before beginning this study. After both arrived, participants were seated separately from one another and provided with an informed consent letter that they both voluntarily agreed to before data collection could begin. Once agreed to, participants completed a number of pre-surveys about their general technology usage.

Then, the experimenter opened up a tutorial and had both participants individually complete a tutorial for Rocket League. This tutorial taught basic controls, and then both participants were given a 5-minute solo training block where they were able to further learn the controls and train in ball movement, passing, and scoring. After these five minutes, each participant played two practice games against a singular AI goalie so they could learn and train in the timed gameplay aspects of Rocket League.

Once these initial training games were completed, participants were introduced to the full task that they would be completing. Participants were told that they were going to play 3 games of Rocket League in a human-AI team, with both participants playing on a team with a different number of AI teammates each game. After each round, participants then completed a set of identical surveys for both their human and AI teammate(s). The order of these rounds was randomized, but as a reminder, only data from the round where participants worked with one AI teammate was

used for analysis. After completing all rounds, participants then participated in a short interview that discussed the perceptions they formed during each round of gameplay.

3.5 Quantitative Data Collection & Measurement

Quantitative data collection consisted of a list of surveys that participants completed after finishing a round of Rocket League with one AI teammate. Participants completed each survey below for both their other human teammate and their singular AI teammate. This analysis resulted in two datapoints being collected per participant (AI and human) and, in turn, four data points being collected for a team as a whole.

The surveys collected for this work relied on the Human-Machine Interdependence Scale [78]. This scale is the first-of-its-kind to itemize specific team processes that comprise and describe interdependence as it relates to human-machine teams. This measurement inventory provides interdependence scales for each of these processes. Together, the multiple metrics collected through this survey tool provide a multifaceted perception of how humans perceived the teamwork they had with their human and AI teammates. The following details each metric that comprises this tool and was measured.

3.5.1 Perceived Conflict. The first measure discussed will be for the perceived conflict participants felt between their other human teammate and their AI teammate. Perceived conflict was measured via four 5-point Likert-style questions. A sample question from this survey includes "This teammate prefers a different outcome than I do in this situation." A greater score for this metric denoted that participants felt that their teammate worked against or hindered their actions during the completed round. In regards to understanding human-AI teamwork, conflict is critical to understanding whether teammates are disruptive to existing team processes. Should this conflict exist, it may not speak to relational issues between teammates but rather struggles in their ability to develop interdependence quickly.

3.5.2 Perceived Mutual Dependence. Participants also rated their perceived mutual dependence between themselves, their other human teammate, and their AI teammate. Mutual dependence was measured via six 5-point Likert-style questions, with a sample question including "We need to work together to manage this situation." A greater score for this metric denotes that participants felt that they and their teammate both relied on each other to ensure a successful team outcome. Mutual dependence is a fundamental aspect of team interdependence, as teammates need to feel that there is an equitable share of interdependence between teammates. Within human-AI teams, mutual dependence might be more complex to develop, as the asynchronous origins of humans and AI might yield a natural barrier or bias that prevents interdependence from feeling mutual.

3.5.3 Perceived Power Over Teammates. Perceived power over one's human teammate and AI teammate was also measured via 4 5-point Likert-style questions. A sample question included "Who felt they had the most influence on the action that was taken?" A greater score denotes that participants felt they had a greater influence over the success of their shared task and resources in comparison to their teammates. Within teams, power is fundamental to interdependence, as teams need to feel as if they can influence the interdependence that their teammates develop. Within human-AI teams, this power may be more difficult to perceive, as the black-box nature of AI and its computational complexity may obfuscate the influence humans can perceive in comparison to their human teammates.

3.5.4 Perceived Future Interdependence From Teammate to Self. Participants completed a survey that denoted the future interdependence that their teammates would have with them. Through four 5-point Likert-style questions, participants rated how strongly they believed their current

interactions with their human and AI teammates would influence the future actions that their teammates would take to interact with them. An example question includes "My behavior in this situation affects how my teammate will interact with me in the future." A higher score would denote that participants felt their teammates would more substantially change their behaviors based on current interactions. Broadly speaking, future interdependence would strongly relate to aspects of coordination in teams, which are a key behavioral result of healthy team interdependence. Within human-AI teams, future interdependence might be different across humans and AI due to difficulties in simultaneously balancing how to be interdependent with both a human and an AI.

3.5.5 Perceived Future Interdependence From Self to Teammate. Participants also completed a survey on perceived future interdependence between themselves and their teammates. This measure explored participants' beliefs that their current interactions with their human and AI teammates would influence the future actions they themselves would take to interact with their teammates. This measure was collected via four 5-point Likert-style questions, such as "The behavior of my teammate in this situation affects how I will interact with my teammate in the future." A higher score denotes that a participant believed that their current interactions would have a stronger influence over their future interactions. Similar to the prior measure, future interdependence from one's self to their teammate might also be a challenge within human-AI teams, as the same behavior might yield two different adaptations due to differences between teammates. This split may burden the coordination capabilities that humans are able to develop.

3.5.6 Perceived Information Certainty From Teammate to Self. Participants also rated the degree to which they felt their teammates understood the beliefs, preferences, and goals of the participant. This measure, known as perceived information certainty from one's teammate to themselves, was measured via five 5-point Likert-style questions. A sample question includes "My Teammate knows why I prefer a certain action." A higher score would signal that participants felt that their human or AI teammate had a strong understanding of the participant. Lastly, information certainty represents one of the strongest interdependence team processes, which is shared understanding, as teammates need to understand the *why* behind teammate actions to truly be independent. In human-AI teams, information certainty could be impacted by the cognitive differences between humans and AI, as humans might not be able to merge the understandings they have for their human teammates and their AI teammates.

3.5.7 Perceived Information Certainty From Self to Teammate. Perceived information certainty was also measured from the participant to their teammates. This measure explored participants' belief that they themselves had a strong understanding of their teammates' beliefs and plans. An example question includes "I know why my teammate prefers a certain action." A higher score denotes that participants felt they had a stronger understanding of how their teammate reasoned through and approached the task they completed. Similar to the prior measure, splitting one's certainty between two types of teammates would likely create two different shared understandings. In human-AI teams, the black-box nature of AI might change whether humans are able to truly be certain whether human and AI teammates have a similar understanding of one another.

3.6 Qualitative Interview & Analysis

In addition to quantitative measures, a short qualitative interview was conducted to expand upon the results of this study. Specifically, this process simultaneously interviewed both human participants, having them take turns to answer each question independently. This interview followed dialogic and phenomenological techniques that focused on the task participants had just completed and how their prior experiences may have influenced this task. Two broad topics were explored during

this interview. First, this interview expanded upon the quantitative metrics explored, with the goal of determining how participants approach teamwork differently with their AI teammate compared to their human teammate. Afterward, the second portion of the interview explored why they likely approach this teamwork differently, exploring the influence of prior experiences, their general beliefs about AI, and any potential interactions they had during the task.

These interviews were transcribed and anonymized for analysis, and a thematic analysis process was followed [7, 10]. Multiple researchers performed a preliminary analysis based on the research questions proposed by this work, and a full analysis was performed after researchers convened and discussed their initial analysis. The result of this process was the creation of two broad themes that expand upon the quantitative results discussed and highlight what factors influenced the perceptions of teamwork that participants formed. Representative quotes were then found for these themes and provided in the results section.

4 Quantitative Results

Due to the design of this experiment, data points possessed multiple layers that needed to be accounted for during analysis. Specifically, the use of a within-subjects factor (AI or Human) meant that correlations at the participant level needed to be accounted for, and the inclusion of multiple human teammates per team meant that correlations at a team level also needed to be accounted for. As such, analysis for this work employed a multi-level modeling (MLM) analysis technique, which explored the difference between perceptions of AI and human teammates while also accounting for correlations between a single participant's multiple ratings and a single team's multiple participants. Prior to model creation, each dependent measure was tested for skewness using D'Agostino's skewness test. In the event that a dependent variable showed significant skewness, a logarithmic correction was applied prior to analysis. Once skewness was shown not to be present or was corrected, a null model was created. This null model was designed with two random intercepts, one for the team number and another for the participant number associated with each data point. After, another model was created that contained both of these random intercepts and also contained the teammate type manipulation (Human or AI). Both models were then compared via a Type III sum of squares ANOVA comparison. The following results section will report the ANOVA results for each of these comparisons, provide post-hoc effects for any significant ANOVA results, and provide graphs that depict significant effects.

4.1 Perceived Conflict

Analysis of perceived conflict explored whether participants perceived significantly different levels of conflict between themselves and their AI teammate and themselves and their human teammate. Significant skewness was found for perceived conflict, so a logarithmic correction was applied prior to analysis. The MLM analysis revealed that, when compared to the null model, the effect of teammate type was found to be nonsignificant, meaning no significant differences were found between the types of teammates participants rated conflict for ($\chi^2(1) = 1.74, p = .187$). These results highlight that neither AI nor human teammates significantly disrupted the formation of teamwork processes and task completion during this study.

4.2 Perceived Mutual Dependence

Mutual dependence was also analyzed using the MLM technique to explore whether participants perceived greater mutual dependence between themselves and their AI teammate or themselves and their other human teammate. A non-significant skewness test revealed that no correction needed to be applied to mutual dependence data. Results of the comparison between the null model and the teammate type model showed that the inclusion of teammate type led to a significant difference

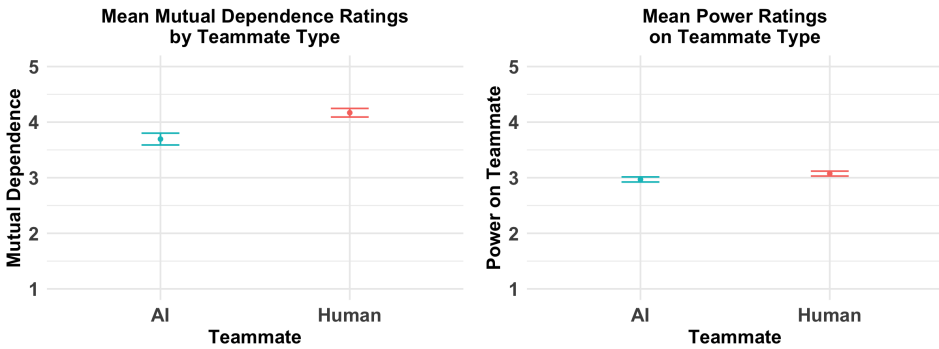


Fig. 1. Perceived Mutual Dependence (left) and Power (right) for human and AI teammate. Error bars denote standard error.

between both models, meaning that the effect of teammate type was found to be significant on the perception of mutual dependence ($\chi^2(1) = 17.88, p < .001$). A post-hoc showed that participants perceived significantly greater levels of mutual dependence with their human teammate compared to their AI teammate ($\beta = .14, SE = .03$). The model's means of both condition levels also show greater perceived mutual dependence between human teammates ($M = 1.42, SE = .03$) than between human and AI teammates ($M = 1.27, SE = .03$). Descriptives for each condition are displayed in Figure 1. Notably, this effect does not mean that participants did not feel like they depended on their AI teammates. Rather, insights from this effect highlight that participants felt dependence was much more mutual between both themselves and their other human teammate, meaning that both teammates depended on each other more equitably. This would imply that teamwork between humans and AI in this task was less equitable or symmetrical, leading to more one-sided relationships.

4.3 Perceived Power Over Teammates

Analysis was conducted to explore whether participants perceived greater influence and power over their AI teammate or their other human teammate. The skewness test was not violated, so no corrections were applied. The MLM analysis showed a significant difference between the null model and teammate type model, meaning there was a significant difference between participants' perceptions of power over their AI teammate and their other human teammate ($\chi^2(1) = 4.25, p = .039$). Post-hoc analysis was performed on the teammate type model for the teammate type effect, showing that participants perceived significantly more power over their other human teammate compared to their other AI teammate ($\beta = .11, SE = .05$). The model's means confirm this analysis, with the mean perceived power on human teammates ($M = 3.07, SE = .05$) being higher than the same for AI teammates ($M = 2.97, SE = .05$). Descriptive means are graphically shown in Figure 1. This result highlights that participants felt their own actions were more influential over their other human teammates, compared to their AI teammates. During the formation of teamwork, it is critical that the actions of individual teammates are influenced by one another, so this effect highlights that the machine nature of AI teammates may lead human teammates to feel less confident that AI will adapt around them, compared to other human teammates.

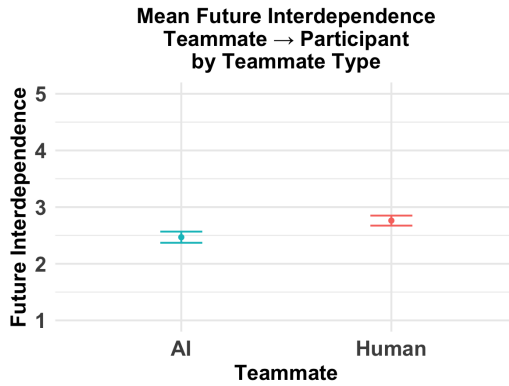


Fig. 2. Perceived Future Interdependence from Teammate to Self for human and AI teammate. Error bars denote standard error.

4.4 Perceived Future Interdependence From Teammate to Self

Perceived future interdependence was analyzed to identify whether participants felt their future actions and decisions would have greater interdependence with their human or AI teammate due to their actions during the study. No significant skewness was found, so no correction was applied. The MLM analysis revealed a significant difference between the null model and the teammate type model ($\chi^2(1) = 5.75, p = .017$). A further analysis of the teammate type model's post-hocs showed a significantly greater level of future interdependence from human teammates compared to AI teammates ($\beta = .29, SE = .12$). This finding is further supported by the model's means of future interdependence from teammates, which are higher for human teammates ($M = 2.76, SE = .09$) compared to AI teammates ($M = 2.47, SE = .09$). Descriptive means have been calculated and displayed in Figure 2. These findings highlight that humans expected themselves to improve in their interdependence with human teammates more so than AI teammates. In turn, asymmetries can still be seen, where humans are likely to grow and develop interdependence at a deep level, yielding two distinct levels of interdependence in human-AI teams.

4.5 Perceived Future Interdependence From Self to Teammate

A similar analysis was performed for the perceived interdependence that participants expected their human and AI teammates' actions to have toward themselves. No significant skewness was found, so no correction was applied. However, the MLM analysis showed that there was no significant difference between the null model and the teammate type model, meaning no significant effect of teammate type was found on perceived interdependence from participants to their teammates ($\chi^2(1) = 0.07, p = .795$). Interestingly, this non-significant result highlights a potential asymmetry in future interdependence. While participants did expect themselves to have significantly different levels of interdependence based on their AI teammate, this difference was not found for the interdependence of teammates. In turn, while participants themselves expect to react differently to human or AI teammates, they may expect both human and AI teammates to react similarly to them.

4.6 Perceived Information Certainty From Teammate to Self

Finally, information certainty was also explored. Starting with the certainty that participants felt in how their understanding of their teammates should inform their own decision-making, skewness was not violated, so a correction was not applied. The MLM analysis and ANOVA comparison

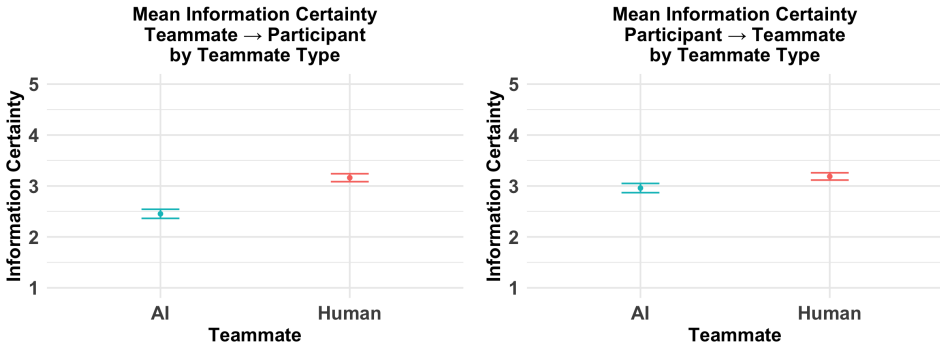


Fig. 3. Perceived Information Certainty From Teammate to Self (left) and From Self to Teammate (right). Error bars denote standard error.

of the null model and the teammate type model showed a significant difference between the two models ($\chi^2(1) = 62.09, p < .001$). In turn, this analysis can identify a significant effect of teammate type on information certainty from teammates to oneself, with post-hocs showing information certainty to be higher from other human teammates than from AI teammates ($\beta = .71, SE = .09$). This finding is further explained through the model's means of each condition, which also show greater levels of information certainty for human teammates ($M = 3.16, SE = .08$) compared to AI teammates ($M = 2.45, SE = .08$). Figure 3 displays the descriptives for each condition. This result highlights that participants likely felt they formed a greater understanding of how they would adapt around their human teammates compared to their AI teammates. This finding would imply that humans would have a stronger grasp of teamwork between other humans, compared to AI systems.

4.7 Perceived Information Certainty From Self to Teammate

Participants' certainty of how their own information influenced the thoughts and decision-making of their teammates was the final test performed. The skewness test was reported to be nonsignificant, so no significant amount of skewness was down for this dependent variable. Therefore, no correction was applied for analysis. For the MLM analysis, there was a significant difference between the null model and the teammate type model for information certainty from oneself to their teammate ($\chi^2(1) = 10.37, p = .001$). This result signals a significant difference between information certainty from oneself to their AI teammate and human teammate, which post-hocs showing a greater level of information certainty for human teammates ($\beta = .23, SE = .07$). Further, model means for human perception ($M = 3.19, SE = .09$) and AI perception ($M = 2.96, SE = .09$) also describe this trend. Descriptive means and standard errors are shown in Figure 3. This finding somewhat reflects the prior finding above. Indeed, participants also felt that their other human teammates had a greater ability to understand and react to the participant. Alternatively, the AI teammate would have a weaker understanding of how . Notably, this effect reflects the participants' perception of information certainty, so AI teammates may or may not have had a more accurate understanding of the participants, but this would not have been reflected in the participants' perception.

5 Qualitative Results

The quantitative results demonstrate a consistent and general gap between the perception of human and AI teammates, despite sharing a number of similarities in their interaction and capabilities. This

gap highlights that AI teammates consistently had lower perceived interdependence than other human teammates across all dimensions of teamwork measured. Specifically, these lower levels were most commonly identified in the mutual dependence, power, and information certainty that participants formed for both AI and human teammates. Thus, to better understand the gap between human-human and human-AI teamwork, it is also important to understand how participants recognize and operationalize this gap. The following qualitative section leverages interview data collected from participants to better understand how and why these gaps might form. In doing so, the following sections separately explore the constructs of, first, information certainty, and, second, mutual dependence and power. For each of these themes, each quoted participant is quoted twice, once to highlight their perception of AI teammates and again to highlight their perception of human teammates. In doing so, this work highlights the real-time gap in participants' development in their perceptions and approaches toward teamwork with AI and humans.

5.1 Humans Create Different Strategic and Understanding Models for Human and AI Teammates, Leading to Different Information Certainty

Firstly, the following theme details how a portion of this gap is most likely represented by the understandings participants created for both their other human teammates and their AI teammates during their team task. Relating back to the quantitative results, this gap is highly reflective of the information certainty construct, as quotes will demonstrate how different levels of certainty can be achieved for AI and humans simultaneously. In turn, this certainty is likely to guide the assumptions and adaptations humans make around their AI and human teammates.

Specifically, past work has identified that the mental models for human and AI teammates are different [67], which is reaffirmed by a gap in information certainty, but the qualitative analysis of this study reveals that the structure of these models might also be different. Ultimately, humans may approach and reason through AI and human interactions differently, leading to non-overlapping information certainty about their AI and human teammates, even if those participants occupy the same role. Thus, a gap between human and AI teammate information certainty would be formed, as shown in the quantitative results. Participant P66 was a clear example of how these understandings differ, providing a vivid description of differing certainty for human and AI teammates:

P66, 18, White, Non-Binary

I feel like the way that I understand a human teammate is fundamentally different from the way I would try to understand an AI. Because I know that AIs are programmed to do a certain thing and that they have formulated actions by design. And so I would never, once I realized that that's what the program was, I never would have anticipated that my AI might take a different route.

I might expect my human teammate to try something different, like maybe based on the response position, or maybe she might have adapted to the different things that AI was doing. And I understood I understand both, but it's a different type of understanding that has to take place.

However, the mere existence of this gap does not explain why AI teammates were often placed lower than human teammates in terms of certainty and even future interdependence. Further exploring this theme, it was revealed that humans actually created more shallow levels of certainty in AI teammates. Often, these interpretations consisted of the actions AI teammates could and did perform, but humans often did not attend to their underlying motives and methods. As a result, the understanding humans created for AI teammates was somewhat lacking in deeper meaning

and explanation. Not only is the above quote from participant P66 an example of this, but the following quotes from P56, who directly compared the understandings they formed for their AI and human teammate. Despite observing similar behaviors being repeated, P56 found it much easier to understand their human teammate:

P56, 19, White Female

Yeah, I say the human teammate, because after they do the same thing over and over, it's easier to predict.

I don't really think I could predict what they [the AI] were doing. I know that their goal was to score because they were on our team. So it was to score against the goalie, but I couldn't necessarily predict their actions, just that they were gonna go towards the ball.

In comparison to the understandings participants created for other human teammates, the ones made for AI teammates were often more complex due to an inability for the human to predict the intermediate actions the AI would take toward the understood, common goal. Ultimately, the shared existence of humans created a level of natural, information certainty and future interdependence where participants assumed that the motives and methodologies of their human teammates would be similar to their own. Thus, if there are gaps in one's understanding of another human teammate, then one can fill in those gaps without actually gaining that understanding. As a result, participants naturally created information certainty and future interdependence with their human teammates, but this was not possible for their AI teammates. The following quotes provide clear examples of how humans see the creation of human-human certainty compared to human-AI certainty as easier due to these reasons:

P03, 18, White, Male

It's kind of hard to predict what the AI is thinking.

But, you can predict what humans are thinking.

P50, 19, White, Female

... I would assume her plan, so then I would think okay, she's assuming my plan.

I just knew that the AI teammates were just trying to win and not really care about teamwork.

P21, 18, White, Female

I feel like I always knew that the AI was just gonna just go for it.

I feel like maybe my human teammate would have more of a strategy.

These differences are most apparent when participants acknowledged that humans and AI shared similar goals but not similar motives, as evidenced by P56's above quote. Ultimately, the sharing of a similar goal is not enough to create a strong perception of information certainty between participants and AI. Rather, it is the perception of a shared experience between humans that allows them to create a perceived certainty of each other, even if that certainty is not grounded in reality but in assumption. Thus, it may be difficult to close the general gap created by these differences, as humans may always relate more heavily to other humans simply due to the nature of their existence. In turn, this shared similarity likely led participants to also share perceptions of strong

teamwork, specifically information certainty and future interdependence, and additional efforts would be needed to improve teamwork between humans and AI.

5.2 Humans have Stronger Expectations for their AI Teammates than their Human Teammates, Leading to Greater Perceptions of Mutual Dependence and Power Between Humans

Importantly, the different understandings humans achieved during interaction were not the only factors creating the general gap identified between human-human and human-AI perception. Additionally, it was found that participants formed much more performance-oriented expectations of teamwork with their AI teammates compared to their human teammates, with human expectations often being more centered around effort. Specifically, humans often felt that their AI teammates were going to be highly performative, but they often did not form any strong expectations for their human teammates' performance. This gap best serves to explain the lower levels of mutual dependence and power humans performed, as these expectations often centered around dynamic human relationships and static AI design. As such, the general gap discovered within human perception may actually exist before interaction even begins, as these expectations translate into beliefs about one's teammates. The following two quotes by participants P09 and P30 illustrate the dichotomy between the formation of these two expectations.

P09, 18, White, Male

My expectations were just that my [human] teammate would give a valiant effort, like give a good effort, go for the ball, try to score and not just sit there and mess around. So like, my expectations with that were definitely met. But like, when there was one bot, it was kind of just like, they were kind of useless.

P30, 18, White, Female

Yeah, I kind of just expected them [AI teammate] to be like good at the game, but they're like robots created to be good at this.
[When asked what expectation they had for their human teammate] Try our best, and [human teammate] exceeded expectations and [redacted] actually scored.

Ultimately, these stronger, performance-based expectations for AI teammates were not driven by any teammate-specific information but rather by fundamental expectations of AI systems. Given that the task completed was in a digital environment, participants felt that the AI systems were naturally skilled at the task. As such, participants somewhat diminished the role that their own actions and influence would play in the development of the AI's own capabilities and actions, likely leading to drops in perceived mutual dependence and power. Specifically, participants who were less skilled created a fundamental understanding that the AI teammates were already more trained and skilled at the task than they were. Importantly, humans were not wrong in this assumption, as the AI teammates have been heavily trained in the environment; however, it is interesting that this mentality not only existed going into the task but also seems to have informed their perceptions of mutual dependence and power. The following quote by participant P68 illustrates the generally higher expectations formed for AI teammates prior to interaction:

P68, 18, White, Male

They know what they're doing a lot better than we do, because they were made for that express purpose of put ball in goal.

[When asked about their expectations for their human teammate] I mean, okay, if I'm being like, flat out honest... not much. I expected myself to be better, but then I realized I haven't played this game in six months. I suck.

More interestingly, however, is that similarly strong expectations were formed by highly skilled players, but those expectations tended to be more negative. Often, these negative perceptions are driven by past experience with AI teammates, which participants often noted as being fairly unskilled. While these perceptions were different from unskilled participants, they were similarly strong with these skilled participants often having fairly clear expectations for their AI teammates, even if their future reality does not meet those expectations. In turn, these strong and negative expectations may also see lower levels of mutual dependence and power, as participants may not have felt these teamwork aspects in prior interactions with AI teammates. Participants P56 and P52 both noted that they had previous experience in Rocket League, and the following quotes are representative of the expectations they had for their AI teammates:

P62, 19, White, Male

I knew that like, since it is an AI bot, like it's not going to really care that much, because it kind of already has like a set plan of what's going to happen. So if you disrupt it a little bit like it just goes out and does it again.

We didn't really try to disrupt... what we [each human teammate] were doing individually, because we just kind of assumed that our teammate is not going to really let us down. So we kind of just like, let one person do their thing, and then we'd kind of be back up if they needed it.

P52, 18, White, Male

When you said a teammate, I assumed it was going to be like the Rocket League AI because Rocket League has their own AI and they suck... One of them hit like a crazy shot, like a really good shot. And I was like dude, what? I was like, I didn't know, they made AI for this game that was like this, like good. But yeah, my expectations were low. And they were definitely better than I expected them to be.

I assumed that you [their human teammate] would be pretty good and I felt like you met that, I felt like we had good teamwork in our skills each were good enough for team play.

However, while the above clearly shows why a gap might exist in general, it is still important to highlight why both skilled and low-skilled players were able to share similar perceptions of mutual dependence and power. For low-skilled participants, this lower perception potentially comes from an expectation that the AI teammates will not need help from them, as the AI is highly skilled and the human is not. Alternatively, higher-skilled participants, who often had past experience with AI teammates, expected AI teammates to be simpler tools. As such, while the familiarity participants had with a domain ultimately determines *why* they form an expectation, the result of said expectation is similar in that neither group expects to have strong and positive teamwork with their AI. This resulted in largely independent work between the human and AI team members, leading to an inability to be mutually dependent or influence on another. Participant P06, who had some experience in Rocket League, and participant P12, who had no experience in Rocket League, provided the following quotes that illustrate this finding:

P12, 18, Black, Female

I guess basically... with AIs, they have more control because they know how to play. With the human, y'all are adjusting and get to learn together.

P06, 18, White, Male

I actually kind of expected them [the AI] to act, how they acted ... the same way and did not have any variants in their strategy, like how they're all just like working the same way. Kind of acting independently from us.

Um, I thought [human teammate] was gonna be really good at this game because [human teammate] played it before. So that was basically it.

The above illustrates that regardless of the skill level of a human, they often find a reason to create strong expectations for AI teammates. In regard to teamwork (i.e., mutual dependence and power), this also means that humans would potentially form expectations around the concept as well. For instance, low-skilled participants would not have expected themselves to be able to contribute strongly to teamwork with AI teammates, and high-skilled participants would not have expected their AI teammates to be able to contribute strongly to teamwork. As a result, a gap between power, mutual dependence, and information certainty was formed, as quantified in this study. This gap may inadvertently bias human teammate perception, ultimately preventing healthy teamwork with humans and AI simultaneously.

6 Discussion

At the core of this study, results show an empirical gap between the strength of teamwork that participants perceive between themselves, their other human teammates, and their AI teammates. If unaccounted for, this gap would likely yield negative consequences to teams, which could impact their performance and social relationships. Fortunately, the qualitative results of this work highlight two potential avenues through which this gap can be addressed. The following discussion explores these potential impacts as well as potential approaches to closing this gap, all of which are synthesized into direct recommendations for human-AI teams.

6.1 Potential Impacts of Teamwork Gaps Between Human and AI Teammates

The most predominant finding of this work comes from the quantitative analysis, which highlights that participants had significantly worse perceptions of teamwork with their AI teammates compared to their other human teammates. Notably, these perceptions still formed despite human and AI teammates having the same role, capabilities, and modalities of interaction. The result of this is, in turn, a potential and inherent gap in the teamwork that humans perceive between their AI teammates and their human teammates. This gap was further identified and explored through qualitative results, which highlight how humans build different understandings for their AI teammates and their other human teammates. Notably, this gap is not entirely new, as prior work has shown that humans perceive and trust AI differently than other humans [22, 23]; however, identifying and observing this gap here highlights that teamwork itself may inevitably look different between these different types of teammates.

Notably, this gap is likely not just going to impact the perception of teamwork but also teamwork itself. For example, cooperative behaviors have been shown to differ when humans work with AI compared to other humans [3, 65]. Similarly, humans can have varying levels of appreciation and aversion toward their AI collaborators based on the context and form of the AI [27, 55]. Generally, these differences are related to the perceived strengths and weaknesses between humans and AI, as

AI brings computational capabilities that afford speed and accuracy while humans excel with their capacity for innovative approaches [84]. The results of this current study further highlight these differences, with participants having seen the computational capabilities that AI brings to the table. As such, it is worth noting that while this work found lower perceived teamwork between humans and AI compared to humans and humans, this difference may not be inherently problematic. Rather, this difference is likely a representation of how participants approached teamwork with humans and AI differently.

Moving forward, it is going to be critical that researchers make an effort to understand, predict, and measure the different types of teamwork in human-AI teams. Based on the qualitative results of this work, traditional human-human teamwork may still have an important place in human-AI teams, with results showing a continued importance for shared understanding [68] and trust [74]. However, the identified gap and its exploration highlight that different factors may be important to human-AI teamwork, including expectation setting [31, 60] and transparency and explainability [63, 77], all of which would help humans gain a more accurate understanding of their AI teammate. Moreover, while it was not explicitly explored in this work, this consideration may also need to account for AI-AI teamwork in human-AI teams with multiple AI, as the teamwork between AI systems will look starkly different from teamwork with humans [9, 32]. In sum, while this work does present a significant gap between the teamwork perceived with human and AI teammates, this gap may not be inherently bad but rather an important consideration to design for. Indeed, future work should explore how to best consider and design the unique needs of human-human, human-AI, and AI-AI teamwork to create highly effective and viable human-AI teams.

6.2 Addressing Inaccurate Understandings and Expectations of Human and AI Teammates

Within this gap that was identified between human and AI teammates, one of the more pervasive qualitative findings of this work was that the expectations and understandings humans form for their AI teammates predominantly led to negative outcomes in perceived teamwork. Indeed, participants struggled to create a strong understanding of their AI teammate, and they also had negative expectations that likely hindered teamwork. These two issues are likely somewhat related, as one's understanding of a teammate is likely driven by that teammate's compliance with one's expectations [24, 31]. Without having clear and realistic expectations for a teammate or technology, it is difficult to form a meaningful understanding quickly. As such, it is worth further exploring these findings and how expectations and understandings can be better aligned within human-AI teams.

The concept and relationship between expectations and understanding can be further supported by teamwork and technology literature. Within the teamwork space, both expectations and understandings can be conceptualized through the concept of a mental model [16, 37]. Within teams, mental models are critical, as they dictate how teammates understand and interrelate the capabilities, information, and goals of their teammates [30], and a shared mental model can dictate how aligned teammates are in these understandings [48]. Within the human-AI space, these mental models will become critical to ensuring researchers and practitioners can measure violations in expectations [24]. Outside of the teaming space, the technology space has also seen a critical need to explore expectations and understandings. For example, the concept of trust calibration has seen a recent focus surrounding AI technology, as humans need to create an accurate understanding and expectations of trust for AI [50, 85]. When accurate, these expectations and understandings can be critical to the effective use of AI technology [46], but misalignment in these aspects can impact complacency or ineffective usage [25, 53]. Within the human-AI teaming space, both of

these conceptualizations and the utilization of expectation and understanding will be critical, as AI will need to garner teaming and technology expectations.

Moving forward, the human-AI teaming space will need to create an effective way of forming accurate expectations and understandings for AI teammates. From an expectations perspective, human-AI teaming is likely going to need to leverage a combination of AI teammate literacy education [38, 79] and human-AI team training [20, 36]. Training, especially, will need a greater focus in the human-AI teaming domain, as the construct has only received minor explorations [44]. From an understanding perspective, human-AI teams will need to create support mechanisms, likely within AI teammates themselves, that continuously measure the understandings that humans form for AI teammates. In turn, these mechanisms would provide a measurable outcome that AI teammates can adapt around with the goal of improving the formation and accuracy of human teammates' mental models [4]. In approaching expectation setting and understanding formation from these perspectives, human-AI teams should not solely focus on ensuring perfect alignment between the two constructs, as doing so would likely be impossible due to the unique experiences of every human. However, future research and implementation of human-AI teams should create a process through which expectations and understanding can be identified and improved to create more effective teammates and technologies.

6.3 Recommendations for Human-AI Teams

6.3.1 Prior to working with AI teammates, humans' expectations about AI need to be identified and discussed. The first recommendation of the proposed work centers around the expectations humans have when entering into a human-AI team. This recommendation posits that humans should explicitly discuss and potentially modify the expectations they have about human-AI teams. Notably, it is impossible to eliminate humans' expectations; however, the results of this work show that when expectations are flawed, they can often lead to differences in teamwork. While these expectations may exist outside of a singular human-AI team, they do inevitably inject themselves into teamwork. Rather than waiting for these expectations to organically develop through team interactions, it may be best to approach them directly before the team begins working together.

Approaching these expectations could benefit from prior work in trust calibration and expectation setting. Indeed, we do not want humans having universally high expectations, as it could lead to complacency [53], or universally low expectations, as it could lead to doubt [76]; both of which are seen in the results of this work. However, calibrating these expectations in human-AI teams likely requires additional considerations not needed in traditional AI work. For example, human-AI teams will need to address the technical capabilities of AI teammates and the social capabilities of AI teammates, which play different roles in teamwork and team effectiveness [15]. In doing so, humans will be more likely to see both human and AI teammates as meeting expectations, even if the teamwork of humans and AI is unique and different.

6.3.2 Human AI teams should strategize how humans should team with one another and with AI teammates. The second recommendation of the proposed work centers around human-AI teams making intentional plans for how to interact with AI and human teammates. Specifically, this work recommends that humans carve out an explicit time that they devote to strategizing how to interact with one another and their AI teammates. Unlike the prior recommendation, the timing of this recommendation is best served after teams have some initial time to interact and form initial perceptions around AI teammates. This recommendation would provide three key benefits. First, in the event that humans plan to interact with AI differently than humans, this difference would become known and accounted for. Second, if humans plan to approach AI teammates differently from one another, these differences can be overcome and addressed without harming teamwork in

the long term. Third, this opportunity would provide a dedicated time where humans can explore their understanding without worrying about maintaining team performance.

In practice, this strategization period would likely take the form of transition phases, which are time periods in teams where humans can step away from their current task and discuss things at a meta-level. All-human teams have a similar, analytical review of taskwork and teamwork that occurs within the transition phases of team operations: the team "debrief" [57]. This active meta-level discussion of current state, successful actions, and problems has shown a promising 25 percent increase in performance in all-human team research [70, 72], suggesting a significant benefit to the successful adoption of a similar process in HATs. Within human-AI teams, transition phases have seen recent exploration, as they provide an opportunity to calibrate the needs that one might have for their AI teammate [64]. However, this work posits an expansion of these transition phases or the adaptation of the team debrief with the goal of creating explicit and distinct strategies for human and AI teammates. In doing so, the differences between human-AI and human-human teamwork will become more apparent.

6.4 Limitations and Future Work

This work contains a few limitations that should serve to inspire the continued exploration into the differences between human and AI teammates. Most notably, this work did not leverage AI and human teammates that were fully identical, as neither were controlled to behave exactly the same. Rather, this work took a more realistic approach to human and AI teammates, understanding that identity is rarely the only difference between a human and an AI teammate. That said, future work should continue to further explore whether or not identity alone impacts teamwork, perception, and interaction. Further, the proposed work solely explored the perception of teamwork. While quantitative and qualitative data were used to explore these perceptions, future work should leverage these findings to identify and quantify specific behavioral and interaction differences between human and AI teammates. Finally, this work had a generally younger audience, with most participants being under 20 years old. With age being a known factor to influence how individuals interact with technology, future work should expand upon the demographics of human teammates who work with AI. Lastly, this work leverages the Rocket League platform, and while efforts have been made to highlight how results obtained in this platform might generalize to other human-AI teams, the nature of teamwork will always require contextual considerations. As such, the results of this work, which rely heavily on a synthetic esports environment, should be extended to other contexts, domains, and types of teams.

7 Conclusion

This work has analyzed the multi-dimensional teamwork needs of HATs through a comparative lens. The natural inclusion of humans and AI agents presents the need to account for the differences between human-human, human-AI, and AI-AI interdependence. This study initiates this movement by identifying a driving empirical gap between the perceptions of human-human and human-AI interdependent teamwork. There was a distinct, consistent, and lower perception of the teamwork between humans and their AI counterparts, but deeper qualitative analyses identified this gap as a consequence of the improper framing of human-AI teamwork. As such, this work exemplifies the need to present human-human and human-AI teaming as unique and to promote realistic, shared expectations among team members of HATs. By explicitly approaching the expectations for human and AI teammates prior to beginning teamwork, HATs can recalibrate the overestimations and underestimations of team members to better align reliance and presumptions. Furthermore, discussing, designing, and executing teamwork processes on individual human-human and human-AI levels may expose higher performing methods and forge more discrete differences between

these interactions. Overall, this study recognized a difference in human perceptions of teamwork when working with other humans and AI teammates, explored the cause of lower AI perceptions, and suggested promoting human-AI teamwork through realistic expectation setting and explicit differentiation.

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Received July 2025; revised November 2025; accepted December 2025